MD SAFWAN ZAMAN

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SUMMARY

PhD student working on automotive cyber-physical system security, developing lightweight authentication frameworks for CAN bus networks, embedded systems, and machine learning–based modeling attack analysis.

Research Area. Automotive Network Security (CAN Bus), Embedded Systems and Real-Time Security, Modeling Attack Resistance and Adversarial ML, Power Electronics.

Supervisor. *Dr. Mohamed Younis*, Professor and Department Chair, Department of Computer Science and Electrical Engineering, University of Maryland Baltimore County.

EDUCATION

- **Ph.D. in Electrical Engineering**, University of Maryland Baltimore County, USA. *August 2024 - current*
- **B.Sc. in Electrical and Electronic Engineering**, Islamic University of Technology, Bangladesh. *2022*

PROFESSIONAL EXPERIENCE

Research Fellow, *University of Maryland Baltimore County, Maryland, USA* *2025*

- Developed a lightweight PUF-based authentication framework to secure ECU communication in CAN bus automotive networks.
- Designed LFSR-based response obfuscation and ECC-assisted error correction to protect against modeling, replay, and impersonation attacks.
- Implemented and validated the protocol on FPGA and Arduino-based CAN platforms for real-time evaluation.
- Performed machine learning–based modeling attack simulations to assess security robustness and adversarial resilience.
- Conducted latency and overhead analysis to ensure compatibility with real-time automotive constraints.

Teaching Assistant, *University of Maryland Baltimore County, Maryland, USA* *August 2024 – current*

- Assisted with C Programming and Embedded Sytems (CMPE 311) course, Systems Design and Programming (CMPE 310) course, Introduction to Electronic Circuits (CMPE 306) course
- Conducted lab sessions, graded exams and assignments, held office hours.

SKILLS

Programming	MATLAB, C/C++, R, Python
Design/Simulation	Simulink, OpenSim, Proteus, Pspice, PSIM, Mide51, LTspice
Microprocessor/Microcontroller	Arduino, MCP2515
Research Tools	LaTeX, GitHub

NOTABLE GRADUATE COURSES

Wireless Sensor Networks (CMPE 684), Introduction to Machine Learning (CMSC 678), Probability and Random Processes (ENEE 620), Neural Engineering and Instrumentation (ENEE 639), Special Topics in Computer Science (CMSC 691), Digital Signal Processing (ENEE 610), Detection and Estimation Theory I (ENEE 621)

PUBLICATIONS

[1] Shagor, Md Rafid Kaysar, et al. "Studying the Applicability of Swarm Intelligence in Designing Optimized PI Controller for DC-DC Zeta Converter: A MATLAB-Based Approach." Mathematical Problems in Engineering 2022.1 (2022): 5816393. ***Link***

SUBMITTED JOURNAL PAPER

[1] Effective Lightweight Security Framework for CAN Bus Systems (IEEE Transactions on Vehicular Technology)

INDUSTRIAL TRAINING

- **Power system by Bangladesh Power Management Institute**
  – Integral components of a Power Generation Plant, Power Transmission and Distribution, Renewable Energy Plant.
- **Communication System by Robi Axiata Limited**
  – 2G/3G/4G System, Modulation and Demodulation of Signals
- **PLC Training by Olympic Industries**
  – Control and Simulation, Digital and Analog Operation, Switching Operation

RECOGNITION

- **General Secretary**, Bangladesh Student Association, University of Maryland Baltimore County. (2025-2026)
  - Organized Several events.
- **Organizer of Esonance**
  - Festival brought on by IUT EEE.